

DWM - 5 lesson

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- * Explain AWS evolution & significant milestones.

AWS (Amazon Web Services) has grown remarkably since its inception, evolving from basic storage & compute services to comprehensive cloud platform offering over 200 fully featured services.

Evolution:-

1) 2002 - 2005:- The foundations:-

- 2002:- Amazon began considering the concept of web services. The company launched Amazon.com Web Services
- 2004 :- launched Amazon SQS (Simple Queue Service) as one of its first web services.

2) 2006:- The launch of AWS:-
officially launched with 2 key services

- Amazon S3 (Simple Storage Service):- provides scalable, reliable data storage infrastructure.
- Amazon EC2 (Elastic Compute Cloud):- allowed users to rent virtual comp. to run their own applicatⁿ.

3) 2007 - 2009

Several new services

- Amazon Simple DB :- A structured data store service.
- Amazon Elastic Load Balancer (ELB):- to distribute incoming traffic across multiple EC2 instances

200 Amazon Cloudwatch, Amazon EC2, etc,
Amazon RDS (Relational Database Service), AWS Management console.

2010 - 2011:-

Introduction to Amazon Elastic Beanstalk
AWS IAM (Identity & Access Management).

2012 - 2015:-

AWS included Amazon DynamoDB &
Amazon Redshift
Expanded with Amazon Kinesis &
AWS OpsWorks.

Major new services:- AWS Lambda
Amazon Aurora, AWS IoT, etc.

2016 - 2018:

- Amazon Polly:- A service that turns text into lifelike speech.

- Amazon Lex:- build⁹ conversational interfaces using voice & text.

- Amazon SageMaker:- A service for building, training, & deploying ML models.

2019 - 2021:-

Enhanced existing services, such as
AWS Transit Gateway:- To connect VPCs & on-premises networks through a single gateway.

AWS Security Hub:- for automating security checks.

Brief Overview of AWS & its offerings

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key characteristics:-

Scalability, Flexibility, Cost-effectiveness, security, Global Reach.

Primary offerings:-

1) Compute:-

• Amazon EC2:-

provides resizable cloud capacity in cloud. Users can run virtual servers, known as instances, on demand.

• AWS Lambda:-

Enables serverless computing, allowing users to run code in response to events without managing servers.

2) Storage:-

• Amazon S3:-

Object storage service, offers high availability, durability, etc.

Amazon EBS (Elastic Block Store) :-
provides persistent block storage
volumes for use with Amazon EC2 instance.

3) Database :-

- Amazon RDS :-
Managed relational database service
supporting various database engines
like MySQL, PostgreSQL, Oracle, etc.
- Amazon DynamoDB :-
Fully managed NoSQL database
services.

4) Machine Learning :-

- Amazon SageMaker :-
Fully managed service that provides
every developer & data scientist with
ability to build, train, & deploy
machine learning models quickly.
- Amazon Polly :-
Service that turns text into lifelike
speech.

57 Security & Identity:-

- Amazon IAM

Security control access to AWS services & resources for users.

- AWS KMS (Key Management Service).

Easy-to-use, highly available, secure key management service.

AWS

GCP

Azure.

Storage services:-

- Amazon S3
- Amazon EBS

- Google cloud storage
- Google cloud Filestore

- Azure blob storage
- Azure files

Scalability:-

- highly scalable, supports automatic scaling S3 & Glacier scale to exabytes of data

- Highly scalable with multi-regional replication - Cloud Storage scales to exabytes of data

- highly scalable, integrates with Azure services - blob storage scales upto exabytes.

- Pay as-you-go pricing model. pricing based on usage, storage class

- Pay-as-go pricing - sustained use discounts for VMs

- Pay-as-go - reserved instances for discounts.

- S3: Object storage with multiple storage classes.

- Cloud storage:- obj storage with various classes.

- Blob storage:- Obj storage with various access tiers.

- Strong security with encryption at rest and in transit - IAM for access control.

- security encryption at rest & in transit IAM access control, VPC services

- Robust security with encryption at rest in transit. Role based Access control

Factors businesses should consider when choosing betw Azure, AWS, GCP?

1) Cost & Pricing Models:-

- Initial and Ongoing Costs:-

Compare the pricing structures, including on-demand, reserved, and spot instances.

- Free tiers & Discounts:-

Evaluate the free tier offerings, potential discounts for long-term commitments, & cost-sav^g programs.

2) Service Offerings:-

- Breadth & Depth of Services:-

Consider variety and maturity of services offered, including compute, storage, databases, ML.

- Specialised services:-

Look for I-like Advanced AI tools, IoT solⁿ, or blockchain.

3) Scalability

Evaluate the ability to scale services automatically based on demand.

Consider No. of data centers and their geographic distributⁿ to ensure redundancy.

4) Security

Security features: Compare built-in security features like encryption, identity and access management, & network security.

5) Ease of Use & Integratⁿ

User Interface & Mng^t Tools:-

- Assess usability of management consoles, dashboards, etc.
- Determine how well cloud services integrate with your current IT infrastructure & third-party tools.

Microsoft Azure in CC:-

- It is a comprehensive cloud computing platform & service created by Microsoft, providing a wide range of cloud services, including computing, analytics, storage, and networking.
- Implements PaaS & IaaS.

Key Features:-

1) Compute Service:-

- Azure VMs :- provides on-demand scalable computing resources with support for windows & linux operating system.
- Azure App Service:- Enables building & hosting of web apps, APIs, etc

2) Storage Service:-

- Azure Blob Storage:- Object service storage offers high availability & durability.
- Azure Files:- Manages file shares that can be accessed via SMB protocol.

3) Database Services:-

- Azure SQL Databases:-
managed relational database service with built-in intelligence.
- Azure Cosmos DB for MySQL/PostgreSQL:-
fully managed database service for open-source databases.

4) Networks

- Azure Virtual Network: Creates private networks in the cloud, with options for IP address ranges, subnets, & network security policies.
- Azure Load Balancer: Distributes incoming network traffic across multiple VMs.

→ AI & ML:- Analytics & big data:-

Azure Synapse Analytics:-
Integrated analytics service combining big data & data warehousing

Azure HDInsight:-
fully managed Apache Hadoop & Spark clusters for big data processing.

6> AI & ML:-

- Azure ML:- end-to-end ml service for building, training, & deploying models.
- Azure bot Service: Platform for building & deploying chatbots.

* AWS Storage Service Utilizatⁿ?

⇒ AWS Storage Service offer a variety of options for storing data in cloud, catering to diff needs within cloud environments.

1> Amazon S3:-

- most popular & highly scalable & durable object storage service.
- It's ideal for storing any kind of data, from website content & applicatⁿ backups to big data archives.
- S3 is very cost effective for storing data.

2> Amazon EBS:-

- provides block-level storage volumes for use with EC2 instances. EBS is ideal for storing data that requires high performance, such as databases.

- offers SSD-backed vol. for high performance & HDD-backed vol. for intensive workloads.
- Enables data backups.

3) Amazon EFS (Elastic File System)

- offers scalable, shared file system for use with EC2 instances.
- EFS is a good choice for applications that require multiple instances to access same files concurrently.

4) Amazon Storage Gateway:-

- creates hybrid storage solⁿ, allow^g you to seamlessly integrate your on-premise storage with AWS cloud storage.
- useful to back up on-premise data to cloud for faster performance.

5) Amazon Glacier.

- Archival storage, cost effective storage for long-term data archiving and backup.
- provides 99.9999999999 (11 9's) durability for archived data.

Q Explain core functionalities of Amazon EC2.

→ Amazon Elastic Compute cloud is fundamental service provided by Amazon Web Services (AWS).

- offers scalable computing capacity in cloud, allow^g users to run virtual servers (instances) on demand

Here are core funcⁿ:

1) Instance types

AWS EC2 provides a wide range of instance types optimized for diff. use cases, such as compute-optimized, memory-optimized, storage-optimized, & GPU instances.

2) Configurations

Users can choose no. of virtual CPU's, memory, storage, network performance.

3) Scalability & Elasticity

Automatically adjusts no. of EC2 instances in response to changes in demand.

Elastic Load Balancing (ELB) :- distributes incoming traffic across multiple EC2 instances

4) Storage optⁿ:-

Am... Elastic Block Store (EBS):-

- It is a block storage volume that can be attached to EC2 instances.

EFS:- provides scalable file storage.

5) Networking features

Am (VPC):- Virtual Private cloud):-

enables users to launch AWS resources into virtual network that they define.

It provides complete control over network configuratⁿ, includ^g IP address, subnets, route tables, gateways etc.

6) Security:-

IAM:- enables secure access to AWS resources without stat using static credentials.